

# MODEL PAPER OF MATHEMATICS FOR GRADE-11 FOR (ANNUAL EXAM-2026)

## Objective Type

**Time Allowed: 30 Minutes**

**Maximum Marks: 20**

- 1. Note:** Four possible answers A, B, C and D to each question are given. The choice which you think is correct, fill that circle with marker or ink pen in the answer-book. Cutting or filling two or more circles will result in zero mark in that question.

|       |                                                                                                                                                                                |
|-------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| i.    | The multiplicative identity in complex number is:<br>(A) (1, 1) (B) (0, 0) (C) (1, 0) (D) (0, 1)                                                                               |
| ii.   | The sum of all the three cube roots of unity is:<br>(A) 1 (B) 2 (C) 3 (D) 0                                                                                                    |
| iii.  | The domain of the function $f(x) = \sqrt{4-x^2}$ is:<br>(A) (-2, 2) (B) (-2, $\infty$ ) (C) ( $-\infty$ , 2) (D) [-2, 2]                                                       |
| iv.   | What is the maximum value of $f(x) = -2x^2 + 4x + 3$ ?<br>(A) 1 (B) 3 (C) 5 (D) 7                                                                                              |
| v.    | If order of a matrix $A$ is $m \times n$ and order of matrix $B$ is $n \times p$ then order of $AB$ is:<br>(A) $n \times p$ (B) $m \times n$ (C) $m \times p$ (D) $p \times n$ |
| vi.   | If $C$ is a matrix of order is $3 \times 3$ and $k \in R$ is any scalar then $ kC $ is equal:<br>(A) $k C^3 $ (B) $k^3 C^3 $ (C) $k^3 C $ (D) $(kC)^3$                         |
| vii.  | An expression of the form $\frac{p(x)}{q(x)}$ , $q(x) \neq 0$ is called:<br>(A) Partial fraction (B) Rational fraction<br>(C) Irrational fraction (D) Polynomial               |
| viii. | The Arithmetic mean between $3\sqrt{3}$ and $\sqrt{3}$ is:<br>(A) 4 (B) $4\sqrt{3}$ (C) $2\sqrt{3}$ (D) $\sqrt{3}$                                                             |
| ix.   | The next term of the sequence 1, 3, 9, 27, ... is:<br>(A) 30 (B) 60 (C) 81 (D) 99                                                                                              |
| x.    | $0!$ is equal to<br>(A) 0 (B) 1 (C) 2 (D) 3                                                                                                                                    |
| xi.   | The number of diagonals of 6-sided figure is:<br>(A) 9 (B) 18 (C) 20 (D) 35                                                                                                    |
| xii.  | The remainder when $f(x) = x^4 + x^3 + x^2 + 1$ is divided by $x + 1$ is:<br>(A) 1 (B) 2 (C) 3 (D) 4                                                                           |
| xiii. | $\operatorname{Cosec}\left(\frac{\pi}{2} + \alpha\right)$ is equal to:<br>(A) $\cos \alpha$ (B) $\sin \alpha$ (C) $\sec \alpha$ (D) $\operatorname{cosec} \alpha$              |

|        |                                                                                                                                                                                          |
|--------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| xiv.   | The value of $\cos\left(\frac{\pi}{2} + \frac{\pi}{6}\right)$ is:<br>(A) $-\frac{1}{5}$ (B) $\frac{1}{2}$ (C) $-\frac{1}{2}$ (D) $\frac{\sqrt{3}}{2}$                                    |
| xv.    | The range of $y = \tan(x)$ is:<br>(A) $(-\infty, 0)$ (B) $(-\infty, \infty)$ (C) $(-\infty, -1) \cup (1, \infty)$ (D) $(-1, 1)$                                                          |
| xvi.   | $\lim_{x \rightarrow a} x^p$ is equal to:<br>(A) $p$ (B) $a$ (C) $p^a$ (D) $a^p$                                                                                                         |
| xvii.  | The notation used by Newton to denote derivative of $y = f(x)$ is:<br>(A) $\frac{dy}{dx}$ (B) $\dot{f}(x)$ (C) $f'(x)$ (D) $Df(x)$                                                       |
| xviii. | The derivative of $f(x) = x^2 - 3$ at $x=2$ ?<br>(A) 4 (B) 2 (C) 3 (D) 5                                                                                                                 |
| xix.   | What will be the angle between the vector $p$ and $q$ , if $p \cdot q = \frac{3}{5}$ and $ p \times q  = \frac{3}{5}$ is:<br>(A) $30^\circ$ (B) $45^\circ$ (C) $60^\circ$ (D) $90^\circ$ |
| xx.    | The magnitude of a vector perpendicular to both the vectors $u = 2i - 3j + k$ and $v = i + 4j - 2k$ is:<br>(A) $\sqrt{50}$ (B) $\sqrt{150}$ (C) $\sqrt{500}$ (D) $\sqrt{1550}$           |

### Subjective Type (Part-I)

Time allowed: 2:30 hours

Max. Marks:80

2. Write short answers to any EIGHT (8) questions:

$2 \times 8 = 16$

- If  $z_1 = (4, 2)$  and  $z_2 = (3, -1)$ , then find  $\frac{z_1}{z_2}$ .
- Show that:  $i^{n+1} + i^{n+2} + i^{n+3} + i^{n+4} = 0$ , for all  $n \in N$ .
- Find the square root of complex number  $5 + 12i$ .
- If  $\omega$  is an imaginary cube roots of unity, prove that  $\frac{a+b\omega^2+c\omega}{a\omega^2+b\omega+c} = \omega$
- Given  $f(x) = x^3 - ax^2 + bx + 1$ . If  $f(2) = -3$  and  $f(-1) = 0$ . Find the values of  $a$  and  $b$ .
- Find the domain and range of  $f(x) = \sqrt{x^2 - 9}$ .
- Find the inverse of  $f(x) = 3x^2 - 2x + 6$ ,  $x \geq 5$ .
- Solve  $|x^2 + 1| = 5$
- If  $A$  and  $B$  are square matrices of the same order, then explain why in general  $(A+B)^2 \neq A^2 + 2AB + B^2$ .

(x) Find the values of  $x$  if  $\begin{vmatrix} 2 & 1 & x \\ -1 & -4 & -3 \\ x & 1 & 0 \end{vmatrix} = 5$

(xi) If  $A$  is a square matrix of order 3, then show that  $|kA| = k^3 |A|$ .

(xii) Resolve  $\frac{2}{x^2-1}$  into partial fractions:

**3. Write short answers to any EIGHT (08) questions:**

**$2 \times 8 = 16$**

- (i) Which term of the A.P., 3, 8, 13, ... is 123?
- (ii) Find the sum of the first 100 positive integers.
- (iii) Find the 12<sup>th</sup> term of  $1 + i, 2i, -2 + 2i, \dots$ .
- (iv) If 5 is the harmonic mean between 2 and  $b$ , find  $b$ .
- (v) Evaluate  $\frac{9!}{6! 3!}$
- (vi) How many signals can be given by 6 flags of different colours, using 2 flags at a time?
- (vii) How many diagonals and triangles can be formed by joining the vertices of a polygon having 15 sides?
- (viii) When the polynomial  $4x^4 + 2x^3 + kx^2 + 13$  is divided by  $x + 1$ , the remainder is 16. Find the value of  $k$ .
- (ix) Suppose a polynomial regression model  $P(x) = 3x^3 - 4x^2 + 2x - 5$ . If a data point at  $x = -1$  is missing. What should be its predicted value?
- (x) Prove that :  $\sin 210^\circ + \cos 240^\circ + \tan 225^\circ + \cot 225^\circ = 1$ .
- (xi) Prove that:  $\cos(\alpha + 45^\circ) = \frac{1}{\sqrt{2}} (\cos \alpha - \sin \alpha)$ .
- (xii) Express  $\sin 5x + \sin 7x$  as a product.

**4. Write short answers to any NINE (09) questions:**

**$2 \times 9 = 18$**

- (i) Find the periods of  $\frac{1}{2} \sin\left(\frac{3x}{2} - \frac{\pi}{2}\right)$ .
- (ii) Find the maximum and minimum values of  $\frac{3}{2} + \cos\left(x - \frac{\pi}{4}\right)$ .
- (iii) Evaluate  $\lim_{x \rightarrow 3} \frac{x-3}{\sqrt{x}-\sqrt{3}}$ .
- (iv) Evaluate  $\lim_{n \rightarrow +\infty} \left(1 + \frac{4}{n}\right)^n$
- (v) Define divergent sequence.
- (vi) A particle moves along a line such that its position after  $t$  hours is given by:  $s(t) = 4t^2 + 2t + 1$  (in miles). Find the average velocity over the interval  $[2, 5]$ .
- (vii) Find the gradient of the curve  $f(x) = 3x^2 + 2x$  at  $x = 1$ .
- (viii) Find the derivative of  $y = \frac{3}{4}x^4 + \frac{2}{3}x^3 + \frac{1}{2}x^2 + 2x + 5$  w.r.t.  $x$ .

- (ix) Differentiate  $\frac{2x-3}{2x+1}$  w.r.t 'x'.
- (x) If  $\underline{u} = 2\underline{i} + 3\underline{j} + \underline{k}$ ,  $\underline{v} = 4\underline{i} + 6\underline{j} + 2\underline{k}$  and  $\underline{w} = -6\underline{i} - 9\underline{j} - 3\underline{k}$ , then show that  $\underline{u}$ ,  $\underline{v}$  and  $\underline{w}$  are parallel to each other.
- (xi) Find  $t$ , so that  $|2\underline{i} + (t-1)\underline{j} + t\underline{k}| = \sqrt{13}$
- (xii) Find the work done, if the point at which the constant force  $\underline{F} = 2\underline{i} + 5\underline{j} + 3\underline{k}$  is applied to an object, moves it from  $P_1(2, -3, 1)$  to  $P_2(7, 5, 3)$ .
- (xiii) Show that  $|\underline{a} \times \underline{b}|^2 = |\underline{a}|^2 |\underline{b}|^2 - (\underline{a} \cdot \underline{b})^2$

## Subjective (Part-II)

**Note: Attempt any three questions.**

**3 × 10 = 30**

5. (a) If  $z_1 = x + yi$  and  $z_2 = a + bi$ , find  $x, y, a$  and  $b$  such that  $z_1 + z_2 = 10 + 4i$  and  $z_1 - z_2 = 6 + 2i$ . 5
- (b) Solve  $|x^2 - 3x + 2| > 4$  5
6. (a) Show that  $\begin{vmatrix} a-b-c & 2a & 2a \\ 2b & b-c-a & 2b \\ 2c & 2c & c-a-b \end{vmatrix} = (a+b+c)^3$  5
- (b) Resolve  $\frac{x^2 - 10x + 13}{(x-1)(x^2 - 5x + 6)}$  into partial fractions. 5
7. (a) For what value of  $n$ ,  $\frac{a^n + b^n}{a^{n-1} + b^{n-1}}$  is the positive geometric mean between  $a$  and  $b$ ? 5
- (b) How many numbers between 100 and 1000 can be formed by using digits 0, 1, 2, 3, 4, 5 without repetition? How many of them are divisible by 5? 5
8. (a) Prove that  $\sin \frac{\pi}{9} \sin \frac{2\pi}{9} \sin \frac{\pi}{3} \sin \frac{4\pi}{9} = \frac{3}{16}$  5
- (b) Given the function  $f(x) = \begin{cases} 2x+3, & x \leq 1 \\ -x+4, & x > 1 \end{cases}$   
Discuss the limit and continuity at  $x = 1$ . 5
9. (a) Differentiate  $\frac{(\sqrt{x}+1)(x^{\frac{3}{2}}-1)}{x^{\frac{3}{2}}-x^{\frac{1}{2}}}$  with respect to  $x$ . 5
- (b) If  $|\underline{a}| = 3$ ,  $|\underline{b}| = 4$  and  $|\underline{a} + \underline{b}| = 5$ . Find the angle between  $\underline{a}$  and  $\underline{b}$ . 5